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SPCE 5085
Assignment #3

1 — Calculate the system noise temperature of a 2 GHz receiver, given the following: $T_{in} = 20K, T_{RF} = 45, T_m = 450K, T_{IF} = 800K, G_m = 1dB, G_{RF} = 20dB, G_{IF} = 30dB$. See text section 4.3.2 (15 pts); Where T_{in} = Temperature In, T_{RF} = Temperature of RF elements, T_m = Temperature of mixer, T_{IF} = Temperature of Intermediate Frequency, and G_m = gain of mixer, G_{IF} = gain of IF. Reminder: Use consistent units: don't mix dB and linear, convert dB units to linear before using equation 4.20.

Convert from dB to linear:

$$G_m = 10^{1/10} = 1.259$$

$$G_{RF} = 10^{20/10} = 100$$

$$G_{IF} = 10^{30/10} = 1000$$

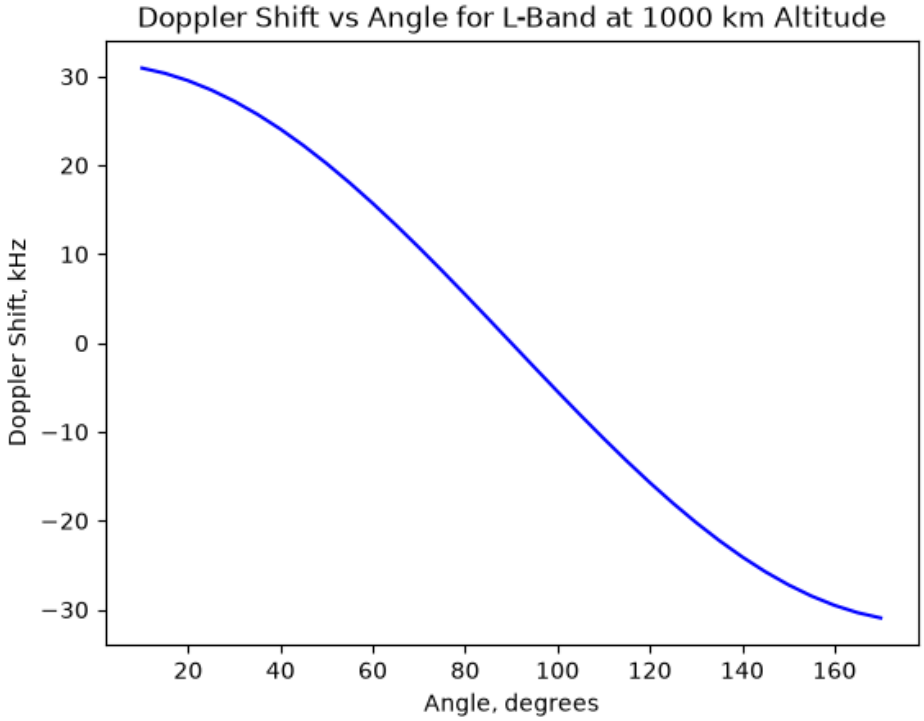
System noise temperature:

$$T_S = T_{RF} + T_{in} + \frac{T_m}{G_{RF}} + \frac{T_{IF}}{G_m G_{RF}}$$

$$T_S = 45 + 20 + \frac{450}{100} + \frac{800}{(1.259)(100)}$$

$$\boxed{T_S \approx 75.85 \text{ K}}$$

2 — Construct a graph showing the Doppler frequency (KHz) for a satellite in LEO (1000 km altitude) transmitting in L band microwave (1.5 GHz) from a ground station at observation angle of 10-170 degrees. A spreadsheet at 5 degree increments works well. (15 pts) Suggest that you read article posted in module 2 on Doppler in LEO.



3 — Construct a complete spread sheet downlink budget, for the Ku-band DBS-TV (Table 4.6a as is in textbook) Inserting all appropriate equations in the spread sheet to reproduce the values. (note typo in table, the path distance units should be 38,000 km (not meters).) (40 pts)

a Is the selection of QPSK at 1/2 rate an appropriate choice? explain

b What would be the CNR and link margin be, if the Transponder was 170 W (vs. 160 W as in table)

4 — Using 14 GHz: A Ku-band satellite uplink has a carrier frequency of 14.0 GHz and a carries a symbol stream at $R_s = 20Msps$. The transmitter and receiver have SRRC filters with $\alpha = .20$

a What is bandwidth occupied by the RF signal?

With matched SRRC filters at transmit and receive, the end-to-end response is a raised-cosine filter, so the occupied RF bandwidth is

$$B = R_s(1 + \alpha)$$

$$B = 20 \text{ Msps} \times (1 + 0.20)$$

$$\boxed{B = 24 \text{ MHz}}$$

b What is the frequency range of the transmitted RF signal?

The signal spans $\pm B/2$ about the carrier:

$$f_c - \frac{B}{2} \leq f_{RF} \leq f_c + \frac{B}{2}$$

$$14.000 \text{ GHz} - 12 \text{ MHz} \leq f_{RF} \leq 14.000 \text{ GHz} + 12 \text{ MHz}$$

$$\boxed{13.988 \text{ GHz} \leq f_{RF} \leq 14.012 \text{ GHz}}$$

5 — Using Figure 5.13 on p.229 of my textbook, estimate graphically:

a the minimum CNR (dB) modulated in BPSK to achieve BER of 10-6 or better.

$\boxed{CNR \approx 11 \text{ dB}}$

b What is the expected BER for a QPSK modulated signal when CNR is about 15 dB

$\boxed{BER \approx 10^{-8}}$